

QIANFENG WEN

* Toronto, Canada

☎ 647-894-7316 ✉ qianfengwen@gmail.com [in LinkedIn](#) [Github](#) [Google Scholar](#)

EDUCATION

University of Toronto

MSc in Computer Science, Supervisor: Ashton Anderson

Sep.2026 –

Toronto, Canada

University of Toronto

Bachelor. Computer Science Specialist, focus on AI

Sep. 2022 – Jun. 2026

Toronto, Canada

GPA 3.91/4.0

PUBLICATIONS (* DENOTES EQUAL CONTRIBUTION)

SafeGEO: Understanding Generative Engine Optimization Risks in Recommendation Agents

2026

Qianfeng Wen*, Yifan Simon Liu*, Xin Liu*, Difan Jiao, Blair Yang, Junda Wu, Zhenwei Tang
arXiv preprint; project page

ThinkTwice: Jointly Optimizing Large Language Models for Reasoning and Self-Refinement

2026

Difan Jiao*, Qianfeng Wen*, Blair Yang, Zhenwei Tang, Ashton Anderson
Colm 2026 under review

Grounded Chess Reasoning in Language Models via Master Distillation

2026

Zhenwei Tang, Qianfeng Wen, Seth Grief-Albert, Yahya Elgabra, Blair Yang, Honghua Dong, Ashton Anderson
Colm 2026 under review

A Simple but Effective Elaborative Query Reformulation Approach for Natural Language Recommendation

2025

Qianfeng Wen*, Yifan Liu*, Justin Cui*, Joshua Zhang, Anton Korikov, George-Kirollos Saad, Scott Sanner
axiv:2510.02656

Natural Language Recommendation via Multimodal Item Scoring Using Gaussian Process Regression with LLM Relevance Judgments

2025

Yifan Liu*, Qianfeng Wen*, Jiazhou Liang*, Mark Zhao*, Justin Cui, Anton Korikov, Armin Toroghi, Junyoung Kim and Scott Sanner
Findings of the Association for Computational Linguistics: ACL 2026

ChessQA: Evaluating Large Language Models for Chess Understanding

2025

Qianfeng Wen, Zhenwei Tang, Ashton Anderson
Colm 2026 Under review

MA-DPR: Manifold-aware Distance Metrics for Dense Passage Retrieval

2025

Yifan Liu*, Qianfeng Wen*, Mark Zhao*, Jiazhou Liang, Scott Sanner
Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP Main)

A Comparative Study of Static and Contextual Embeddings for Analyzing Semantic Changes in Medieval Latin Charters

2024

Yifan Liu, Gelila Tilahun, Xinxiang Gao, Qianfeng Wen and Michael Gervers
COLING'25 Workshop: LoResLM (oral)

Elaborative Subtopic Query Reformulation for Broad and Indirect Queries in Travel Destination Recommendation

2024

Qianfeng Wen*, Yifan Liu*, Joshua Zhang, George Saad, Anton Korikov, Yury Sambale and Scott Sanner
RecSys '24 Workshop: ROEGEN

Monte-Carlo Tree Search for Behavior Planning in Autonomous Driving

2024

Qianfeng Wen, Zhongyi Gong, Lifeng Zhou and Zhongshun Zhang
IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR) '24 (oral)

WORKING EXPERIENCE

ZBot Technology <i>Co-founder & CTO</i> Led development of a Export oriented AI agent system.	October 2025 – Present <i>Toronto, Canada</i>
Air Canada <i>AI Engineer Intern (in collaboration with D3M Lab @ UofT)</i> - Led development of a travel recommender system that adapts to free-form user queries. - Curated and released TravelDest, the first benchmark dataset for travel search, combining WikiVoyage content with insights from travel experts. - Developed an LLM-based query reformulation + dense retrieval pipeline, improving R-precision by 12% over baseline.	June 2024 – August 2024 <i>Toronto, Canada</i>
Bosch <i>Autonomous Driving Intern</i> - Contributed to Bosch’s Wave3 autonomous driving program with a focus on motion planning algorithms. - Designed an MCTS-based decision-making framework for complex urban/highway scenarios; achieved a 98% success rate with planning latency < 0.2s on project benchmarks.	May 2023 – August 2023 <i>Shanghai, China</i>

RESEARCH EXPERIENCE

CSS Lab (University of Toronto) <i>Undergraduate Researcher; Supervisor: Ashton Anderson</i> - Created ChessQA, a benchmark to evaluate large language models on comprehensive chess understanding (tactics, strategy, notation, and explanation quality). - Explored supervised fine-tuning and reinforcement learning to train and distill chess-specialized LLMs for reasoning and move justification.	May 2025 – Present <i>Toronto, Canada</i>
D3M Lab (University of Toronto) <i>Summer Intern & Undergraduate Researcher; Supervisor: Scott Sanner</i> - Built Retrieval-Augmented Generation (RAG) QA pipelines combining sparse and dense retrieval (FAISS) with LLMs; developed LLM-driven query reformulation EQR and user-intent modeling to improve retrieval relevance. - Proposed Manifold Distance, a training-agnostic metric inspired by Manifold Ranking, outperforming cosine/dot-product measures on Dense Passage Retrieval (DPR) benchmarks. - Introduced GPR-LLM: a Gaussian Process Regression approach with LLM relevance judgments to model multimodal item relevance, consistently exceeding DR, cross-encoder, and pointwise LLM rerankers across four NLRec datasets and two backbone LLMs.	May 2024 – March 2026 <i>Toronto, Canada</i>
The Matter Lab (University of Toronto) <i>Undergraduate Researcher & Full-time Summer Intern; Supervisors: Alán Aspuru-Guzik, Haozhe (Marko) Huang, Pan Chen</i> - Explored multi-agent debate (MAD), Chain-of-Thought (CoT), and Tree-of-Thought (ToT) methods to strengthen LLM logical reasoning; analyzed whether LLMs exhibit schema-based learning behaviors. - Investigated compositional training methods (e.g., Ito density sampler) for diffusion models on 2D-ARC problems.	April 2024 – January 2026 <i>Toronto, Canada</i>

PROJECTS

Decision-making in Autonomous Driving using Monte Carlo Tree Search (MCTS) Matlab - Designed and implemented a comprehensive decision-making framework employing MCTS for diverse autonomous driving scenarios. - Successfully validated the framework’s effectiveness in varied conditions, ranging from complex urban intersections to highway exits.	90 Stars on github
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PROFESSIONAL SERVICE

Reviewer, IEEE International Conference on Robotics and Automation (ICRA) 2025

HONORS & AWARDS

2026. Vector Scholarship in Artificial Intelligence. Vector Institute, Toronto. Research. \$17,500
2025. Department of Computer Science Research Award. Department of Computer Science, University of Toronto. Institutional. Research. \$8,000.
2023-2025. Dean’s List Scholar. University of Toronto. Institutional. Academic. \$0.
2024-2025. Chancellor’s Scholarship. Trinity College, University of Toronto. Institutional. Academic. \$500.
2024. Educational Training Stipend. Department of Mechanical & Industrial Engineering, University of Toronto. Institutional. Research. \$2,500.

TECHNICAL SKILLS

Languages: Python, Java, C, Matlab, R

Technologies/Frameworks: verl, vLLM, SGLang, Prompt Engineering (OpenAI, LangChain), Pytorch, SkLearn, GitHub, Matlab Tool Boxes